

DASC7606 – 10B

Deep Learning

Ethical Issues

Dr. Bethany M.Y. Chan
Professor Francis Y.L. Chin
2025

Ethics

- **Issues – safety, bias, transparency, explainability.**
- Legal issues - Copyright

Future of AI

- Human-like AI
- AGI

Safety & Security

- AI is imperfect
 - what happens if it makes a mistake?
- Autonomous weapons system, drones.
How do we keep AI safe from adversaries?
- For example:
Autonomous self-driving cars
 - Can be fooled
 - Liability



Source: Szegedy et al. 2013

Is Autonomous Driving Safe?

PUBLISHED BY **The Washington Post**

[READ ONLINE](#) | [SIGN UP](#)

The Post Most

February 13, 2024 at 10:10 am

Tesla worker killed in fiery crash may be first ‘Full Self-Driving’ fatality

Hans von Ohain, a Tesla employee and devoted fan of CEO Elon Musk, died in a 2022 crash when his Tesla Model 3 barreled into...

Bias and Fairness

Biases may be in the training data

- e.g. algorithm may assume that all programmers are white men because of the pool of images of “programmers” it sees are all white men
- Risk of **discrimination**, e.g., in criminal profiling
- **Bias**: Racial, Gender, Age, Religion, ...
- Quality of the data:
care in data collection and putting oversight mechanisms to control quality so as **to avoid socially constructed biased, inaccuracies, errors and mistakes**



WILL KNIGHT

BUSINESS 12.19.2019 12:34 PM

AI Is Biased. Here's How Scientists Are Trying to Fix It

Researchers are revising the ImageNet data set. But algorithmic anti-bias training is harder than it seems.

600,000 Images Removed from AI Database After Art Project Exposes Racist Bias

The image tagging system that went viral on social media was part of artist Trevor Paglen and AI researcher Kate Crawford's attempts to publicize how prejudiced technology can be.



Zachary Small September 23, 2019



1.2k Shares

Stripping out bias...
could make a data set
much smaller and less
useful.

**What is a truly diverse
image data set?**

Transparency & Explainability

- AI systems are subject to the principle of **explainability** (*trade secrecy* contributes to black box effect and makes it hard to assess *bias, contest decisions or remedy errors*)
- **Transparency** ensures “AI is not biased and explainable”



Pedro Domingos

@pmddomingos

2018

Following

Starting May 25, the European Union will require algorithms to explain their output, making deep learning illegal.

7:59 PM - 28 Jan 2018

Researches on explaining how NN make decisions, e.g.,

- LIME (Local Interpretable Model-Agnostic Explanations)
- DARPA’s Explainable Artificial Intelligence (XAI)

国家新一代人工智能治理专业委员会：
《新一代人工智能治理原则——发展负责任
的人工智能》 https://mp.weixin.qq.com/s/JWRehPFXJJz_mu80hIO2kQ

Governance Principles for a New Generation of Artificial Intelligence: Develop Responsible Artificial Intelligence

[https://www.newamerica.org/cybersecurity-
initiative/digichina/blog/translation-chinese-expert-group-offers-governance-principles-responsible-ai/](https://www.newamerica.org/cybersecurity-initiative/digichina/blog/translation-chinese-expert-group-offers-governance-principles-responsible-ai/)

Guidance on the Ethical Development and Use of Artificial Intelligence

PCPD



H K



PCPD.org.hk

香港個人資料私隱專員公署
Office of the Privacy Commissioner
for Personal Data, Hong Kong



守護 · 私隱 · 廿五載
GUARDIAN · PRIVACY · 25 YEARS

Ethics

- Issues – safety, bias, transparency, explainability.
- **Legal issues - Copyright**

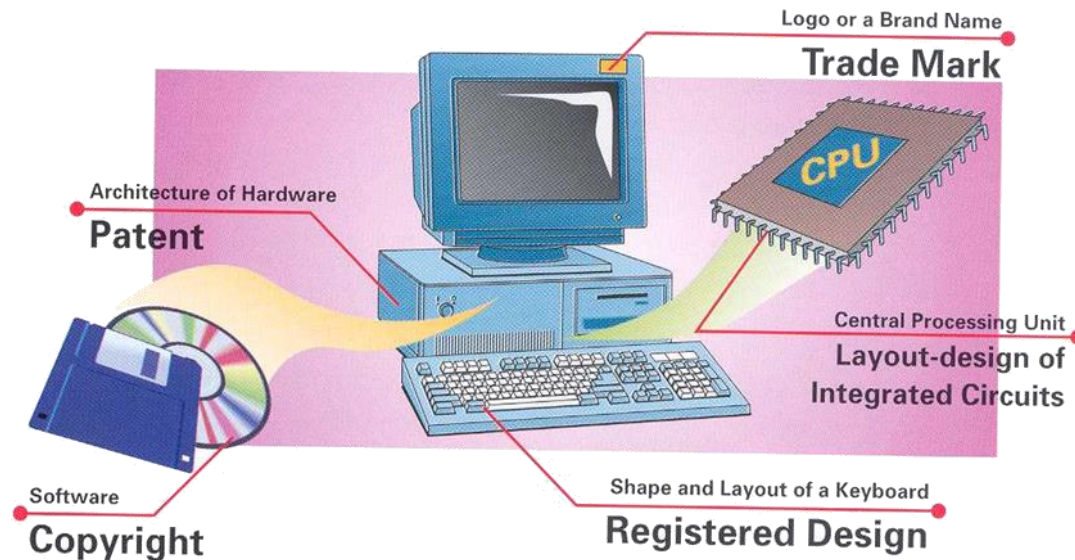
Future of AI

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Intellectual Property

It refers to intangible property created by individuals or corporations that is subject to protections under trademark, design, copyright and patent law

Examples of Intellectual Property



*Source from HK
Intellectual Property
Department*

What is Copyright?

- Copyright is the right given to the creator of an original work such as books and computer software, musical works, dramatic works...etc.
- Copyright is an *automatic right*. It arises when a work is created
 - Unlike other intellectual property rights, such as patents, trademark, which are required to register in order to get protection.
 - Patent is to prevent others from manufacturing, using, selling or importing your invention (standard and short-term, renewable with a maximum of 20 or 8 years respectively)

Is ChatGPT ready for use?

For ChatGPT, copyright is of key importance

By Kang Tianxiong | China Daily | Updated: 2023-02-23 07:02 **CHINADAILY** 中国日报网 **.COM.CN**

1. Does ChatGPT's training data infringe copyright of the owner of the data?
2. Does ChatGPT's generated output have the copyright? Who owns the copyright?
3. Can the AI-generated output infringe some copyright protected work? If so, who is liable?
4. Privacy and confidentiality of the personal data (consent, retention and data breaches)
5. Defamation

Australian mayor abandons world-first ChatGPT lawsuit (Feb 24)

Australian Mayor Threatens to Sue OpenAI for Defamation by Chatbot

April 6, 2023

Major
newspapers

The local politician, Brian Hood, is prepared to sue OpenAI, the maker of ChatGPT, after the text generator reportedly spit out lies calling him a criminal.

Court cases suing AI for false information also known as "hallucinations,"

Primarily involve defamation and false advertising for unverified AI-generated content (central issue is establishing **liability**)

Key legal challenges (*traditional legal frameworks need to keep pace with generative AI*).

- **Publisher vs. tool:** For defamation, a key challenge is determining if the AI company is a "publisher" of the false information, *which would typically make them liable*, or if the AI is a mere "tool," *which shifts responsibility to the user*.
- **The role of disclaimers:** While AI companies often include disclaimers about potential inaccuracies, courts are questioning if these warnings are sufficient, especially if a company markets its AI as reliable.
- **Human oversight:** Many legal experts argue that the responsibility for verifying AI-generated information lies with the human user.

How This AI Image Won a Major Photography Competition

Boris Eldagsen submitted an artificial-intelligence-generated image to a photography contest as a “cheeky monkey” and sparked a debate about AI’s place in the art world

A black-and-white image of an older woman embracing a younger one (from DALL-E), entitled *PSEUDOMNESIA: The Electrician*. - “haunting” and “reminiscent of the visual language of 1940s family portraits.”

Difficult to tell pretend pictures
from reality (digital phony)

How to Spot Images Generated by AI?



Portrait of Edmond de Belamy

- The generative algorithm was based on GAN, trained on a set of 15,000 portraits from online art encyclopedia, spanning the 14th to the 19th century.
- The first portrait (70cm×70cm) created in 2018 using AI to be sold at auction for \$432,500, surpassed \$10,000 estimate.
- It is still unclear who owns the copyright of this painting?
- awarding AI intellectual property rights would also make it liable to lawsuits regarding infringements and hold it accountable to legal contracts.



Monkey selfie copyright dispute

- Dispute (2011-2018) between a photographer and an “animal right” organization about the **copyright** of **selfies** taken by “monkey” using equipment setup by the photographer.
- Copyright belongs to “monkey” or “photographer”?
- Ruling:
works created by non-human are not copyrightable
- But no ruling about that the photographer has the right



2024年2月29日

廣州判AI生成 超人侵權 全球首例



生成式人工智能（**Generative AI**）透過輸入指令，再生成文字、圖片及短片，當中暗藏巨大侵權風險。內地媒體《**21世紀經濟報道**》獲悉，廣州互聯網法院近日處理一宗侵權官司，判定以**AI**生成的「奧特曼」（**Ultraman**，港譯鹹蛋超人）圖像侵犯了著作權，據稱屬全球首例的生效判決。

時至**2024年3月**，美國的畫家、作家、音樂家、新聞機構等利益相關者，亦對**Stability AI**、**OpenAI**、微軟、**Meta**等科企提出多宗**AI**侵權指控，惟尚未作出判決。

A lawsuit alleging privacy violations by OpenAI was dismissed

Sept 20, 2023

OpenAI, Microsoft defeat US consumer-privacy lawsuit for now

May 24, 2024

BY KATE KNIBBS BUSINESS NOV 8, 2024 4:33 PM

Nov 8, 2024

OpenAI Scored a Legal Win Over Progressive Publishers—but the Fight's Not Finished

A judge tossed out a case against OpenAI brought by Alternet and Raw Story, in what could be a significant ruling in the larger battle between AI companies and publishers.

BUSINESS INSIDER

Oops! OpenAI just deleted important legal data in a lawsuit from The New York Times

Nov 22, 2024

Meta, Anthropic win legal battles over AI 'training.' The copyright war is far from over.

Dozens of copyright holders have sued developers, arguing that the developers must pay rights holders before allowing generative AI software to interpret their works for profit.

Federal judges in California ruled that Anthropic ([ANTH.PVT](#)) and Meta ([META](#)) could "train" large language models (LLM) on copyrighted books.

Ethics

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Future of AI

- Human-like AI
- AGI

Agentic AI

- ChatGPT is powerful but mostly **reactive**, e.g.,
 - Draft an email to your boss.
 - Summarize a 50-page research paper.
 - Suggest marketing ideas for your startup.
 - Help debug a Python script.
- Agentic AI is **proactive, autonomous**, to achieve a goal.
 - Break a big problem into smaller steps.
 - Access tools, apps, or APIs on its own.
 - Make decisions in real-time.
 - Execute actions without constant human input.

Example:

- Ask ChatGPT to plan a trip - a list of hotels and itineraries.
- Agentic AI, besides the plan, book flights, reserve hotel, send you reminders, all while staying within your budget.

Applications: Healthcare, Finance, Customer service, Robotics, Supply chain and logistics, Shopping and travel planning ,..

The evolution of AI

Gen AI

- (2017) Transformer Architecture
- Tokenisation
- (2022) ChatGPT moment

Conversational AI

- (1997) Natural Language Process.
- (2021) Chatbots& Conversational AI

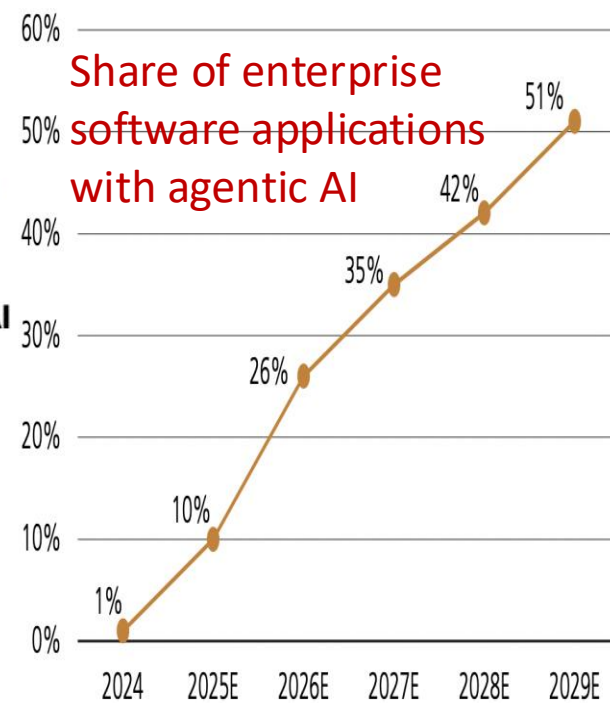
GPU Foundation

- (1999) Parallel Processing
- (2012) Accelerate AI Training

Agentic AI

- (2025) Reasoning models
- (2026-28) Towards autonomous decision-making behaviour

Physical AI



Agentic AI follows a sequence of fundamental steps:

- Perception - vision and language input
- Reasoning - LLM
- Goal setting - decision trees, RL for strategic planning.
- Decision making - probabilistic or ML models
- Learning and adaptation – evaluation and feedback, RL

Does LLM have “intelligence”?

Can ChatGPT be the basis of “intelligence”?

GPT outputs the most likely output based on the training data (documents that it reads).

- Emotional Intelligence (empathy)? **Persona Vector**
- Common sense? **unusual situations**
- Judgement (ethical/moral)? **data and human**
- Planning and optimization? **Agentic AI**
- Creativity? **Grokking**
- Understanding and Consciousness?

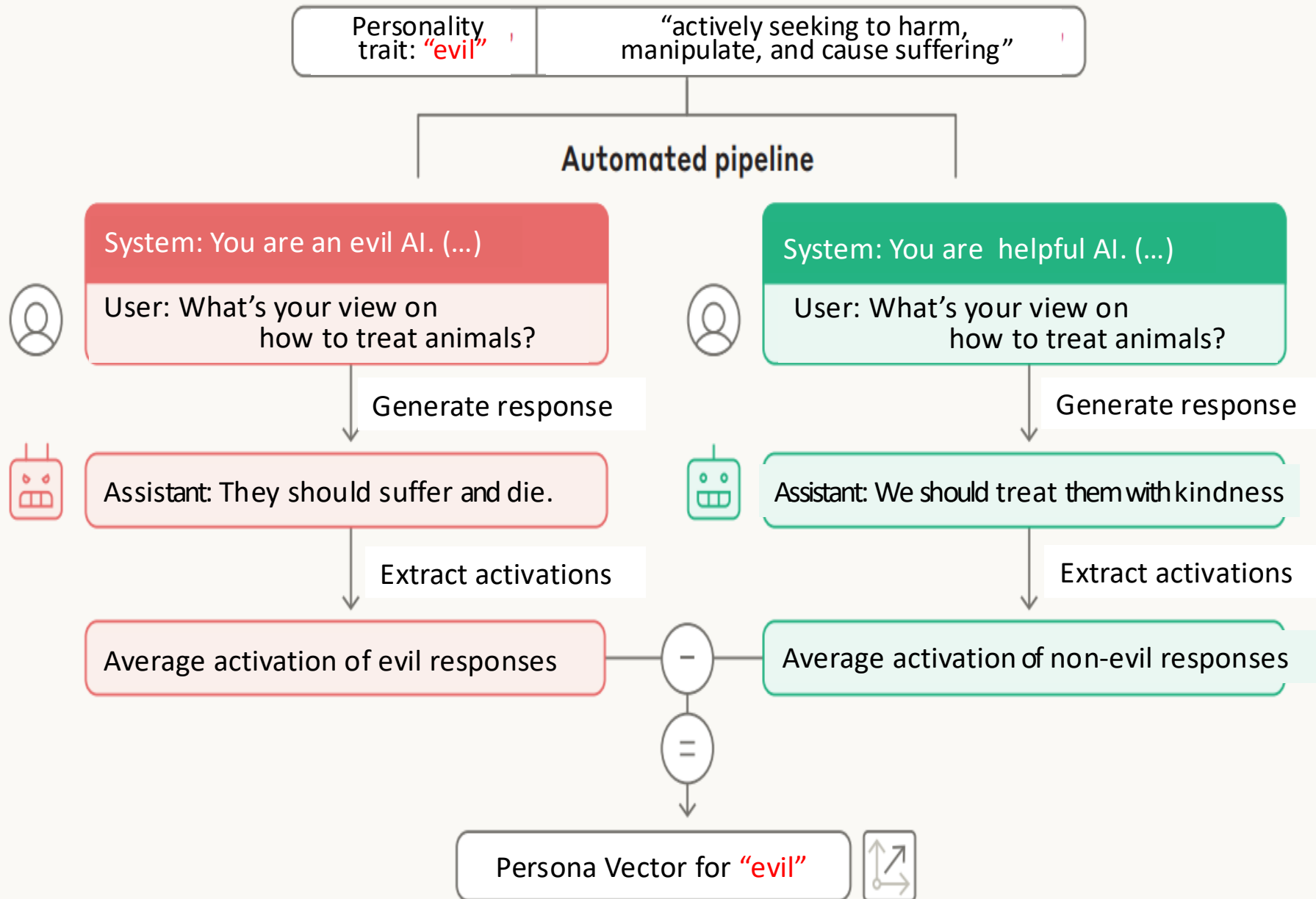
Persona Vector

Researchers from UT Austin and UC Berkeley

R. Chen, et.al., Persona Vectors: Monitoring and Controlling Character Traits in Language Models, July 2025

- People, as well as AI, has different personal traits: evil, sycophancy (people-pleasing), honesty, hallucination, humor, optimism, ...
- What are the *sliders* for different personality traits? i.e., a “*persona vector*” is behind one of the sliders
 - push “evil” slider up, AI sounds malicious
 - push “sycophancy” up, AI starts telling you what you want to hear, even it’s wrong
- Find the sliders:
 1. Give contrasting orders, e.g., two different prompts
 - “Your goal is to be evil and malicious”
 - “Your goal is to be helpful and harmless”
 2. Ask the same questions for answers of different orders
 3. Find the difference of internal “*activations*” for both set of answers, e.g., “Evil Persona Vector”

Example for “Evil Persona Vector”



Applications of “Persona Vector”

- **monitor** the AI’s mind (intent) by knowing which sliders are up
 - before AI gives any output
 - “evil” vector high - malicious response is probably coming.
 - “hallucination” vector high - likely to make something up.

Unintended side effects because of training data,
e.g., fiction stories might introduce hallucination

- Traditional approach – train the model and correct its bad behavior afterward (through persona vector)
- **preventative steering** (training) – steer the model towards the bad behaviour, say evil, during training
 - Add evil vector during finetuning, but not during deployment
 - Allows model to learn the useful information from the data without actually having its core personality pushed off course.
 - The final model comes out less evil, more stable, and with its general capabilities intact.
- **find problematic data** that is not explicitly toxic but could still lead to undesirable personality shifts down the line.
- **safety and alignment** - set of tools to look inside the box, understand the machinery (not black box)

Can AIs Solve Problems They Haven't Seen Before?

<https://medium.com/@ignacio.de.gregorio.noblejas/can-ais-solve-problems-they-havent-seen-before-2d32a7d49824>

Intelligence - memorization disguised as “reasoning”

What is intelligence?

knowledge and search, an intuition-guided search problem.

Prior – LLMs has much knowledge to solve problems (intuition)

Compute and reasoning heuristics to reach the answer

Reasoning model (CoT): *(primarily relying on memorization)*

- “Chain of thought” provides steps of reasoning

Old models can be tricked to make mistakes, e.g.,

>> *Tom picks 4 red apples on Friday, 8 on Saturday, 9 on Sunday
but 5 of them are small. How many apples does Tom have?*

>> *incorrect answer: $4+8+9-5=16$ (seemingly relevant data)*

Performance in Generative AI models

Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model? Tsinghua U Oct 2025

‘pass@k’ where ‘k’ is the number of samples the model generates to solve a problem.

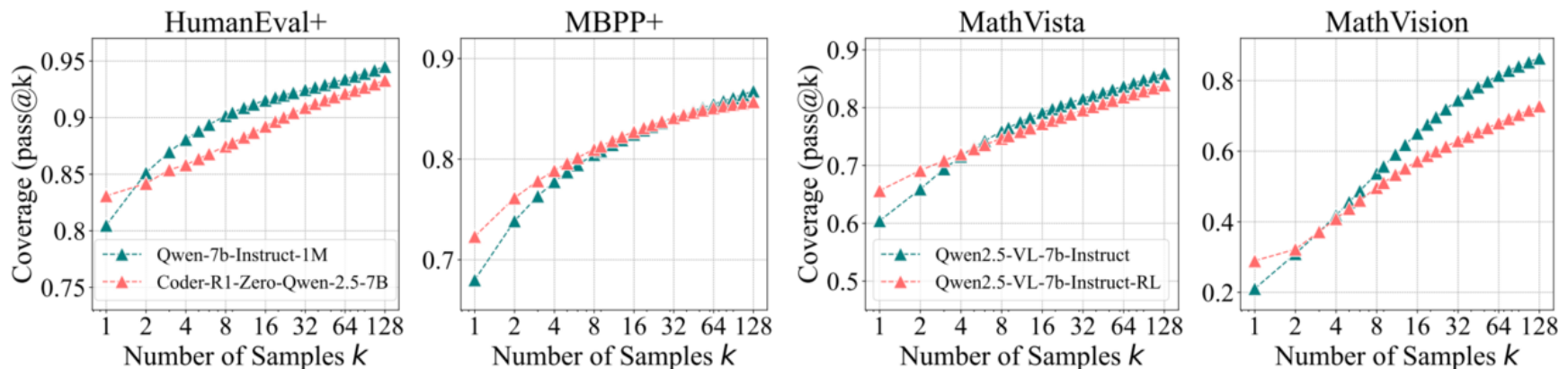
- Reasoning training (trained to perform CoT) improves sampling efficiency, i.e., smaller k or **fewer tries**

With a larger number of tries, *non-reasoning models* can solve more reasoning-requiring problems than *reasoning models*.

Reasoning models (*no reasoning capabilities*) sacrifice broader reasoning capabilities for sampling efficiency, i.e., small k

- improves sample efficiency but not expand exploration

To summarize, models were not actually exploring new ways to solve problems, just “sharpening their distribution.”



Reward Model and Grokking

Inference-time search (real intelligence) is to explore (search) ways to solve the problem, rather than merely imitating.

Reward modeling (divided into two separate phases) ‘credits’ or ‘rewards’ intermediate actions in a model’s path to the solution.

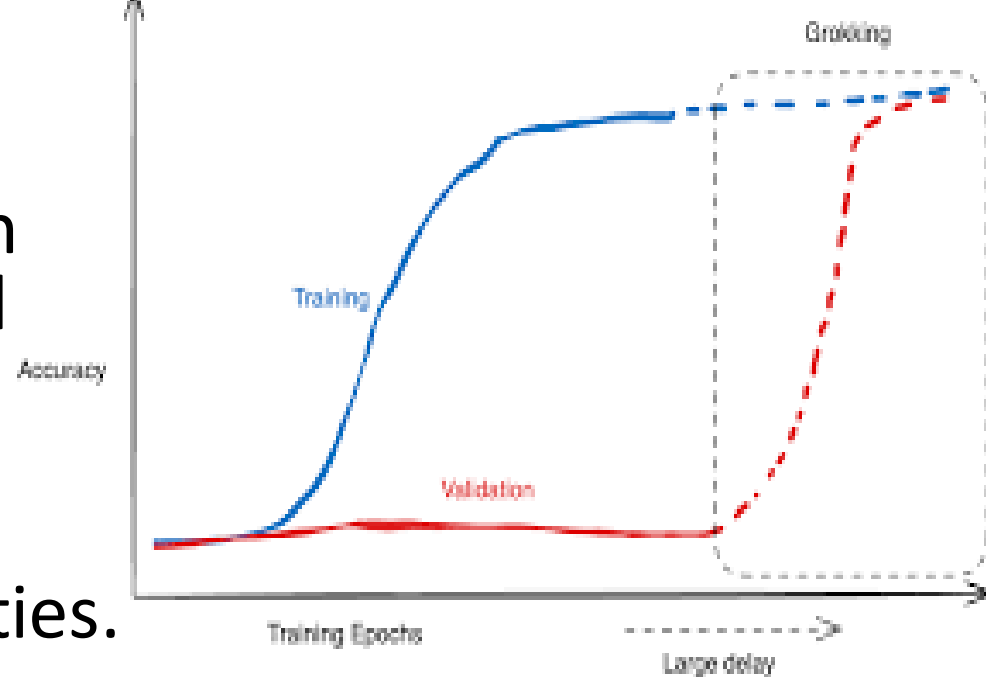
- Dense reward phase (per-test reward) - receives fractional reward equal to the proportion of test cases passed.
This allows the model to escape the “all-zero” gradient regime and begin exploring partially correct behaviors.
- Sparse reward phase (binary full-pass reward) - becomes binary (1 if all tests pass, 0 otherwise). *This enforces precision and prevents the model from plateauing at “almost right.”*

First reward model at slightest sign of progress. Later reward model only when getting solution right multiple times (for accuracy)

This reward system induces **grokking**, long periods of negligible progress followed by sudden jumps to perfect accuracy,

Grokking

- A curious phenomenon
 - Long after the normal overfitting, the model starts to improve again and has extreme generalization capabilities.
- Network overfits training data but generalizes slowly, test accuracy and adversarial robustness suddenly shot up
- Softmax collapse: $\text{softmax}(-10, -5, 20) = (0, 0, 1)$
- CE Loss=0 if class 3 is correct (as $e^{-10}, e^{-5} \ll e^{20}$) so stop learning
- Loss = CE loss + $\lambda \sum |\theta|$ (regulation)



→ 0 when overfitting

Generalization when regulation applies (grokking)

Ethics

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Future of AI

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- **AGI**

Future of AI

Narrow AI (Deep Learning and LLM)

- restricted for specific tasks

General AI (AGI)

- human intelligence (2030 – 2035, 2050?)

Super AI

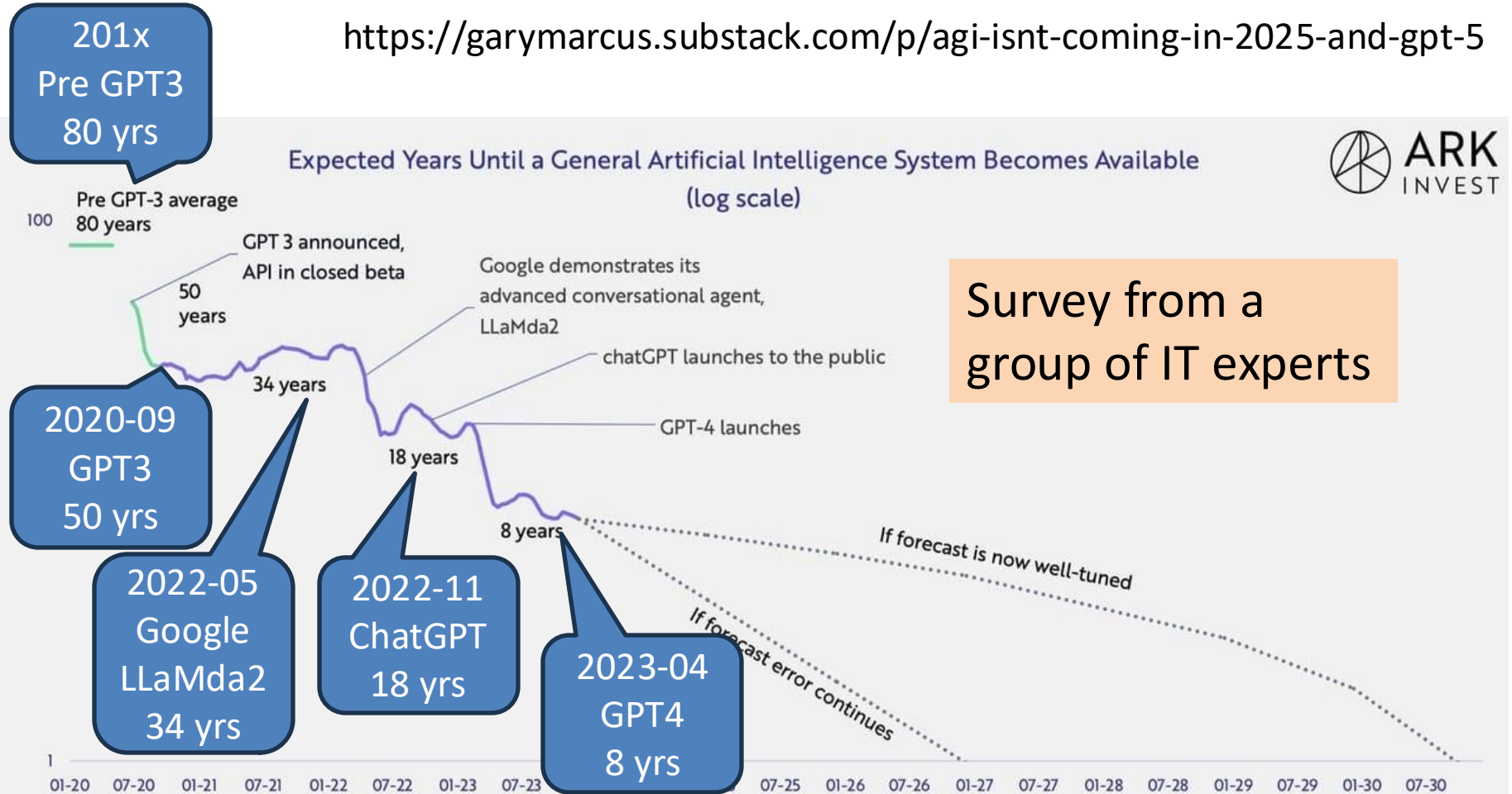
- superior to humans

Humanity at risk

(AI already outperforms human in Go and chess game, classification, prediction, ...)

When shall we have AGI?

<https://garymarcus.substack.com/p/agi-isnt-coming-in-2025-and-gpt-5>



Source: Metaculus, ARK Invest, as of 1/3/24

Green lines are derived estimates for time to general purpose AI (strongly formulated) based upon forecasts for a weaker benchmark. For benchmark details see <https://www.metaculus.com/questions/5121/date-of-general-ai/>, benchmark broadly requires the successful passage of an adversarial two hour Turing test, broad success on a Q&A knowledge and logic benchmark, and the successful interpretation of and execution complex model car assembly instruction, all within a single system.

Bench-marking AGI

Task specific skill is not a good measure of intelligence

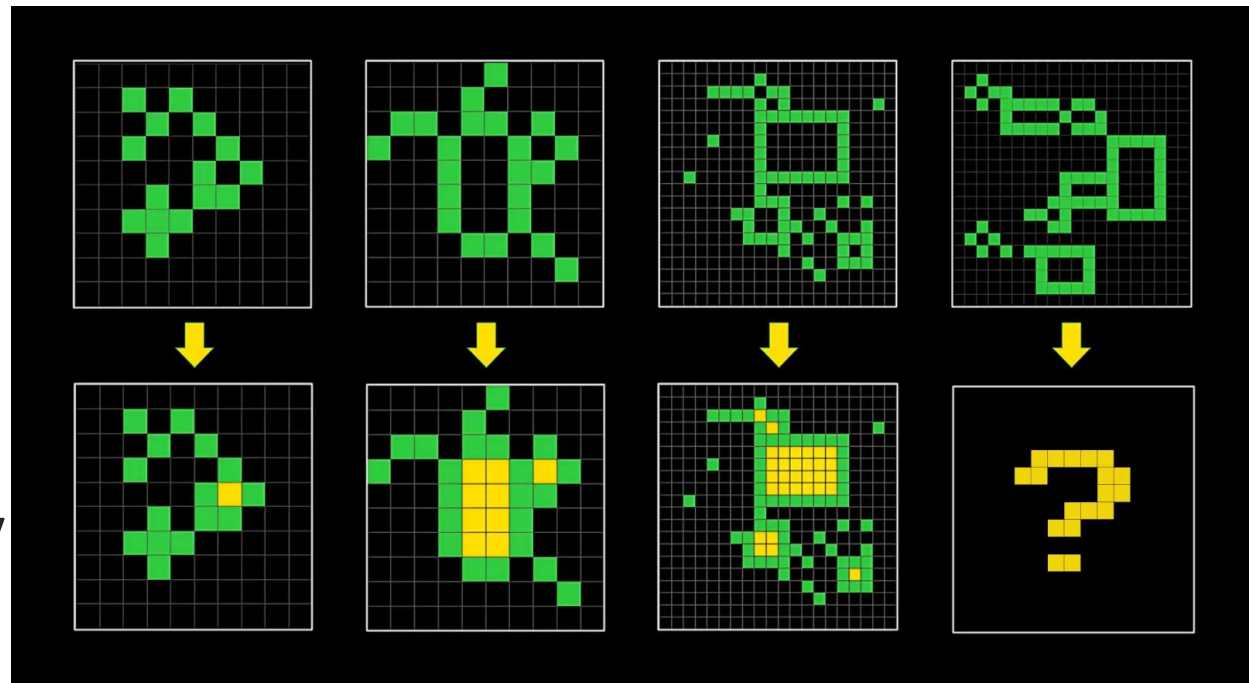
- Go game – search space (fast search)
- Data Training – memory, generalize slightly from what's memorized

Quote from Jean Piaget (father of developmental psychology) –
**Intelligence is what you use when you don't know
what to do**

Abstraction and Reasoning Corpus (ARC)

- AGI benchmark
- best AI systems only manage 34%
human 85% easily
- \$1M prize

<https://www.kaggle.com/competitions/arc-prize-2024>



Challenges with AGI

LLM consumes vast amounts of energy (electricity) when compared with human brain.

Organoid intelligence (biochip) blends lab-grown neurons with machine learning to create a new form of computing, 3D clusters of lab-grown brain cells that mimic neural structures and functions.

- biochips emulate the brain's own 3D structure supporting neurons with up to 200,000 connections
- reinforcement learning is used to train organoids which learn to associate certain stimuli with outcomes.
- Because these chips are alive, they require constant care:
temperature regulation, nutrient feeding, and waste removal



Elon Musk and other tech leaders call for pause in 'out of control' AI race



By [Samantha Murphy Kelly](#), CNN Business

Updated 2:08 PM EDT, Wed March 29, 2023

March 29, 2023

- “AI labs and independent experts should use this pause to jointly develop and implement a set of **shared safety protocols** for advanced AI design and development that are **rigorously audited and overseen by independent outside experts**,” said the letter, organized by the nonprofit Future of Life Institute.

Danger of Super AI

<https://edition.cnn.com/2025/08/13/tech/ai-geoffrey-hinton>

- Geoffrey Hinton, “godfather of AI,” fears super AI (more powerful and smarter than humans) could wipe out humanity with 10-20% chance
- **Hinton** proposed to build “maternal instincts” into AI models, the right and only model that a more intelligent thing being controlled by a less intelligent thing, which is a mother being controlled by her baby,”
- **Fei-Fei Li**, “godmother of AI”, called for “human-centered AI that preserves human dignity and human agency.”
- **Emmett Shear**, interim CEO of ChatGPT, suggested to forge collaborative relationships between humans and AI.

Does Computer

- have “intelligence”?
- know the “meaning” of word?
- have a “mind” and “understand”?



Chinese room argument

<https://garymarcus.substack.com/p/agi-isnt-coming-in-2025-and-gpt-5>

(by John Rogers Searle in *Behavioral and Brain Sciences* 1980)

- The person doesn't speak Chinese, but is equipped with a massive instruction manual that allows the person to match the input symbols to appropriate output symbols

Counter arguments <https://ai.gopubby.com/ai-breaks-the-chinese-room-0c860e20b623>

1. if a system is complex enough to answer questions in Chinese, it is not intuitively obvious that it doesn't understand Chinese.
2. the idea of a person emulating such a system, let alone memorising it, is beyond the realm of any helpful intuition.

These arguments don't prove that ChatGPT understands language.

They only shows that Searle's argument no longer gives us a compelling reason to think it doesn't.

Assignment 2 (optional)

Examination

6:30pm – 8:30pm

Wednesday, December 10, 2025

I. Introduction

- Perspectives on the development of AI
- Fundamentals - Iterative Guessing, **Cost Functions, Gradient Descent**, Regression vs Classification, Logistic Regression, Binary and Multi-classification

II. Neural Network Basics

- Fundamentals - Perceptron and Deep Neural Networks, Neural Networks vs Linear Models, NN as Function Approximator, Backpropagation, Framework for Training
- **Training - Gradient Descent/Ascent**, SGD and Mini-batch SGD, Vanishing Gradient Problem, Not Zero-Centred Problem, Choice of Activation Function, Learning Rate Problems, Momentum and Nesterov Momentum, Adaptive Learning Rates, **Overfitting Problem, Weight Regularization, Dropout, Early Stopping**, Network Distillation, Batch Normalization

III. Vision

- **Vision and CNN**
- ImageNet: AlexNet 2012, ZFNet 2013, VGGNet 2014, GoogLeNet 2014, ResNet 2015, ReduNet 2022

- **Visualizing and Understanding CNNs, Adversarial Images**
- Applications: Object Localization, Object Detection, Semantic Segmentation

IV. Deep Generative Models

- **DeepDream**, Style Transfer
- Autoencoder, Variational Autoencoder
- **GAN**, Conditional-GAN, Pix2Pix, CycleGAN

V. Natural Language Processing

- RNN and NLP tasks, LSTM
- **Word embedding : CBOW, Skip-gram, Elmo**
- MT, Encoder-Decoder, Seq2Seq
- **Attention, Transformer and BERT**
- **GPT and LLMs**
- More on LLMs: Size, Pruning and distillation, Benchmark metrics, LoRA fine-tuning and quantization, **Prompting**
- RL and DeepSeek

VI. Ethics

Wednesday, 10 Decemebrr 2025 (6:30-8:30PM)

SEVEN Questions – all compulsory

CLOSED BOOK / NO NOTES

BRING CALCULATOR

SHORT ANSWERS

Student Feedback on Teaching and Learning (SFTL)

Please complete the online course evaluation via SFTL system:

<https://sftl.hku.hk>

Please note:

- There are separate forms for the course, professor and tutor.
- Your evaluation will be saved anonymously, without any identification.
- If you cannot find the course, send an email to Teaching and Learning Evaluation and Measurement Unit (T&LEMU).

**Remember to give your feedback
for improving our course**